

Environment Configuration Guide

ROSHN Parametric Masterplan Modelling Platform

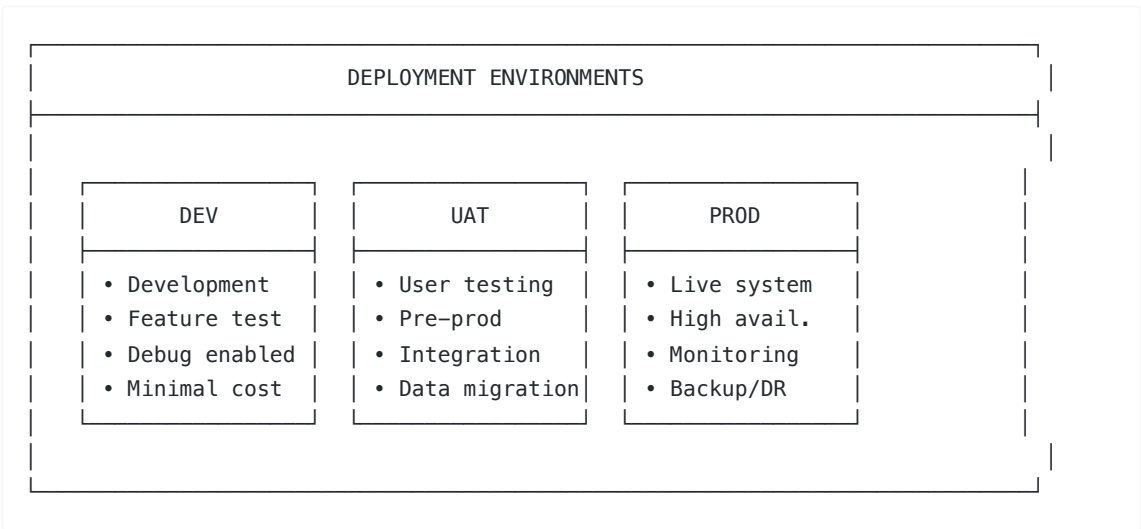
Version: 1.0 **Date:** 5 January 2026 **Status:** Production Ready

Overview

The ROSHN Platform supports three deployment environments, each designed for specific use cases and user groups.

Environment	Purpose	Availability	Users
Dev	Development and feature testing	Business hours	5-10 developers
UAT	User Acceptance Testing	Business hours	10-20 testers
Prod	Live business operations	24/7	50-100+ users

Environment Summary



1. Development Environment (Dev)

Purpose

- Feature development and testing
- Developer debugging and experimentation
- Low-cost, minimal resources
- Not publicly accessible

1.1 On-Premise Specifications

Component	Specification
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CPU	4 vCPUs
RAM	8 GB
Storage	100 GB SSD
OS	Ubuntu 22.04 LTS / RHEL 8+
Network	100 Mbps
Database	PostgreSQL 14+ (same server)
SSL	Optional (self-signed acceptable)
Domain	dev.platform.roshn.sa
Port	3001
PM2 Instances	1

1.2 Azure Specifications

Resource	SKU	Specification
App Service Plan	B1 (Basic)	1 core, 1.75 GB RAM
App Service	Linux	Node.js 20 LTS
PostgreSQL	B_Gen5_1	1 vCore, 2 GB RAM, 32 GB storage
Storage Account	Standard LRS	10 GB

Resource Group: roshn-platform-dev-rg **Region:** UAE North (uaenorth)

Deployment:

```
./scripts/deploy-azure.sh dev
```

1.3 GCP Specifications

Resource	Type	Specification
Cloud Run	-	512 Mi RAM, 1 vCPU
Cloud SQL	db-f1-micro	Shared core, 0.6 GB RAM
Cloud Storage	Standard	10 GB
Min Instances	0	Scale to zero
Max Instances	2	Limited scaling

Project: roshn-platform-dev **Region:** me-central1 (Dammam)

Deployment:

```
./scripts/deploy-gcp.sh dev
```

2. UAT Environment (User Acceptance Testing)

Purpose

- User acceptance testing before production
- Pre-production validation
- Integration testing
- Data migration testing
- Stakeholder reviews

2.1 On-Premise Specifications

Component	Specification
CPU	4 vCPUs
RAM	8 GB
Storage	150 GB SSD
OS	Ubuntu 22.04 LTS / RHEL 8+
Network	1 Gbps
Database	PostgreSQL 14+ (separate server recommended)
DB CPU	2 vCPUs
DB RAM	4 GB
DB Storage	100 GB SSD
SSL	Required (valid certificate)
Domain	uat.platform.roshn.sa
Port	3002
PM2 Instances	2
Load Balancer	Optional

2.2 Azure Specifications

Resource	SKU	Specification
App Service Plan	S1 (Standard)	1 core, 1.75 GB RAM
App Service	Linux	Node.js 20 LTS
PostgreSQL	GP_Gen5_2	2 vCores, 4 GB RAM, 128 GB storage
Storage Account	Standard LRS	50 GB

Resource Group: roshn-platform-uat-rg **Region:** UAE North (uaenorth)

Deployment:

```
./scripts/deploy-azure.sh staging
```

2.3 GCP Specifications

Resource	Type	Specification
Cloud Run	-	1 Gi RAM, 1 vCPU
Cloud SQL	db-custom-1-3840	1 vCPU, 3.75 GB RAM
Cloud Storage	Standard	50 GB
Min Instances	0	Scale to zero
Max Instances	5	Moderate scaling

Project: roshn-platform-uat **Region:** me-central1 (Dammam)

Deployment:

```
./scripts/deploy-gcp.sh staging
```

3. Production Environment (Prod)

Purpose

- Live business operations
- 24/7 availability
- High availability configuration
- Full monitoring and alerting
- Regular backups and disaster recovery

3.1 On-Premise Specifications

Component	Specification
Application Servers	2x for high availability
CPU (per server)	8 vCPUs
RAM (per server)	16 GB
Storage (per server)	250 GB SSD (RAID 1)
OS	Ubuntu 22.04 LTS / RHEL 8+
Network	1 Gbps (redundant)
Database Server	Dedicated

DB CPU	4 vCPUs
DB RAM	16 GB
DB Storage	500 GB SSD (RAID 10)
DB Replication	Standby recommended
SSL	Required (valid certificate)
Domain	platform.roshn.sa
Port	3000
PM2 Instances	max (all CPUs)
Load Balancer	Required
Backup Storage	1 TB

3.2 Azure Specifications

Resource	SKU	Specification
App Service Plan	P1V2 (Premium)	1 core, 3.5 GB RAM
App Service	Linux	Node.js 20 LTS, 2 instances min
PostgreSQL	GP_Gen5_4	4 vCores, 8 GB RAM, 256 GB storage
Storage Account	Standard GRS	100 GB
Azure CDN	Standard	Static asset delivery
Application Insights	-	Monitoring and logging

Resource Group: roshn-platform-rg Region: UAE North (uaenorth)

Deployment:

```
./scripts/deploy-azure.sh prod
```

3.3 GCP Specifications

Resource	Type	Specification
Cloud Run	-	2 Gi RAM, 2 vCPU
Cloud SQL	db-custom-2-4096	2 vCPU, 4 GB RAM
Cloud Storage	Standard	100 GB
Cloud CDN	-	Static asset delivery
Cloud Monitoring	-	Logging and alerts

Min Instances	1	Always running
Max Instances	10	Auto-scaling

Project: roshn-platform **Region:** me-central1 (Dammam)

Deployment:

```
./scripts/deploy-gcp.sh prod
```

Environment Comparison

Resource Comparison

Resource	Dev	UAT	Prod
App CPU	4 vCPU	4 vCPU	8 vCPU x 2
App RAM	8 GB	8 GB	16 GB x 2
App Storage	100 GB	150 GB	250 GB x 2
DB CPU	Shared	2 vCPU	4 vCPU
DB RAM	2 GB	4 GB	16 GB
DB Storage	50 GB	100 GB	500 GB
Network	100 Mbps	1 Gbps	1 Gbps x 2
Load Balancer	No	Optional	Yes
SSL	Optional	Yes	Yes
High Availability	No	No	Yes
Backup Retention	7 days	14 days	30 days

Cloud SKU Comparison

Environment	Azure App	Azure DB	GCP Run	GCP SQL
Dev	B1	B_Gen5_1	512Mi/1CPU	db-f1-micro
UAT	S1	GP_Gen5_2	1Gi/1CPU	db-custom-1-3840
Prod	P1V2	GP_Gen5_4	2Gi/2CPU	db-custom-2-4096

Domain Configuration

Environment	On-Premise Domain	Port
Dev	dev.platform.roshn.sa	3001

UAT	uat.platform.roshn.sa	3002
Prod	platform.roshn.sa	3000

Environment Variables

Each environment requires the following configuration:

Variable	Description	Dev	UAT	Prod
NODE_ENV	Environment mode	development	production	production
DATABASE_URL	PostgreSQL connection	Dev DB	UAT DB	Prod DB
NEXTAUTH_SECRET	Auth encryption key	Unique	Unique	Unique
NEXTAUTH_URL	Application URL	Dev URL	UAT URL	Prod URL

Security Configuration

Feature	Dev	UAT	Prod
HTTPS/SSL	Optional	Required	Required
2FA	Optional	Recommended	Required
IP Restrictions	No	Optional	Recommended
VPN Access	No	Optional	Recommended
Audit Logging	Basic	Full	Full
Session Timeout	24h	8h	4h

Monitoring & Alerting

Metric	Dev	UAT	Prod
Health Checks	Manual	Hourly	Every 5 min
Log Retention	7 days	14 days	30 days
Alerts	None	Email	Email + SMS
Uptime SLA	None	95%	99.5%

Backup Configuration

Setting	Dev	UAT	Prod
Database Backup	Daily	Daily	Daily

Backup Retention	7 days	14 days	30 days
Point-in-Time Recovery	No	No	Yes
Geo-Redundancy	No	No	Recommended

Deployment Commands

Azure

```
# Development
./scripts/deploy-azure.sh dev

# UAT
./scripts/deploy-azure.sh staging

# Production
./scripts/deploy-azure.sh prod
```

Google Cloud Platform

```
# Development
./scripts/deploy-gcp.sh dev

# UAT
./scripts/deploy-gcp.sh staging

# Production
./scripts/deploy-gcp.sh prod
```

On-Premise

```
# Development
sudo ./scripts/deploy-onprem.sh dev

# UAT
sudo ./scripts/deploy-onprem.sh staging

# Production
sudo ./scripts/deploy-onprem.sh prod
```

Related Documentation

Document	Description
Azure Deployment	Complete Azure deployment guide

GCP Deployment	Complete GCP deployment guide
On-Premise Deployment	Complete on-premise guide
Infrastructure Sizing	Detailed sizing specifications
High-Level Design	System architecture overview